Remote Repository

- A remote repository is a GIT repository stored on another system or on an online repository hosting platform. It allows developers to store, share, and synchronize project history with other developers.

- A local repository is stored on the developer's computer, while a remote repository is stored separately and can be accessed through a network.

Local Repository vs Remote Repository

Local Repository	Remote Repository
Stored on the developer's computer.	Stored on another system or repository hosting platform.
Can be used without continuously connecting to a network.	Usually accessed through a network connection.
Contains project files and GIT history.	Can contain a shared copy of project files and GIT history.
Used for local development and commits.	Used for sharing and synchronizing work.

Why Remote Repositories are Used

- To store a shared copy of a project.
- To back up project history outside the local computer.
- To share code with other developers.
- To synchronize work between developers.
- To support collaboration using branches and commits.
- To make a project accessible from multiple development environments.

Understanding Remote

A remote is a reference or name that represents a remote repository. The commonly used default remote name is origin.

```
origin  →  Remote Repository
```

The name origin is only a convenient name. A different remote name can also be used.

Connecting a Local Repository to a Remote Repository

Before pushing local commits to a remote repository, a local repository must be connected to the remote repository.

Basic Workflow

```
Local Repository
      ↓
Add Remote
      ↓
Remote Repository
      ↓
Push Changes
```

Adding a Remote Repository

Command

```
git remote add origin <repository-url>
```

Purpose of the Command

- Adds a remote repository and gives it the name origin.

Example

```
git remote add origin https://github.com/username/student-management.git
```

Here, origin is the remote name, and the URL identifies the remote repository.

Note

- The URL must point to the intended remote repository.
- The remote name origin is commonly used, but it is not mandatory.

Viewing Remote Repositories

Command

```
git remote -v
```

Purpose of the Command

- Displays the configured remote repository names and their URLs.

Example

```
git remote -v
```

```
origin  https://github.com/username/student-management.git
(fetch)
origin  https://github.com/username/student-management.git
(push)
```

Pushing Changes to a Remote Repository

Command

```
git push origin main
```

Purpose of the Command

- Sends local commits from the specified branch to the specified remote repository.

Example

```
git switch main
git push origin main
```

Here, origin is the remote name and main is the branch being pushed.

Pulling Changes from a Remote Repository

Command

```
git pull
```

Purpose of the Command

- Gets changes from the configured remote tracking branch and integrates them into the current local branch.

Example

```
git switch main
git pull
```

This is commonly used when the remote repository contains changes that are not yet present in the local branch.

Fetching Changes from a Remote Repository

Command

```
git fetch
```

Purpose of the Command

- Download new data and updates remote-tracking information without automatically integrating the changes into the current working branch.

Example

```
git fetch
git status
```

Fetch is useful when a developer wants to inspect remote changes before deciding how to integrate them.

Cloning a Remote Repository

Command

```
git clone <repository-url>
```

Purpose of the Command

- Creates a local copy of an existing remote repository, including its project files and GIT history.

Example

```
git clone https://github.com/username/student-management.git
```

A new local project directory is created from the remote repository.